BK Restaurant Management System

Software Requirements Specification

Course	SENG 31242 – System Design Project
Group	01
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Document Status	Final

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1. Introduction

1.1 Purpose and Scope

The purpose of this document is to specify the complete software requirements for the BK Restaurant Management System. This system is designed to digitize and centralize operations for BK Restaurant and BK Bar across three branches located in Ambalangoda and Kahawa, Sri Lanka.

The scope of this system encompasses:

- A custom Point of Sale (POS) system for counter order management and billing
- A proprietary mobile application for delivery riders with live GPS tracking
- Automated inventory management with low-stock alerts and supplier notifications
- A centralized web dashboard for multi-branch monitoring and analytics
- An employee management module covering attendance, shift scheduling, and leave applications

This system directly replaces the current manual operations that rely on WhatsApp messaging, physical ledger books, and unmonitored delivery assignments.

1.2 Definitions, Acronyms, and Abbreviations

Definitions	
Restaurant Owner	The primary user who manages all three branches, oversees analytics, and approves business decisions.
Employee (Reception / Kitchen)	A supporting user who handles walk-in orders at the POS counter, prepares food based on real-time kitchen displays, and manages their attendance.
Delivery Rider	An operational user who receives delivery assignments on their mobile device and navigates to customer locations using GPS.
Supplier	A supporting external user who receives digital inventory requests and confirms stock availability.

Acronym	
POS	Point of Sale – the digital counter system used to process orders and generate bills.
GPS	Global Positioning System – used for live rider location tracking.
SRS	Software Requirements Specification
KDS	Kitchen Display System – a screen in the kitchen showing incoming orders.
UC	Use Case – a description of system interactions with actors.
FR	Functional Requirement – numbered FR-01, FR-02, etc.

Acronym	
NFR	Non-Functional Requirement – system quality attributes.
SME	Small and Medium Enterprise.
REST API	Representational State Transfer Application Programming Interface.
JWT	JSON Web Token – used for secure session authentication.

1.3 References

- Aggarwal, S. (2018). Modern Web Development with React. Packt Publishing.
- Boysen, N., Fedtke, S., & Schwerdfeger, S. (2021). Last-mile delivery concepts: a survey from an operational research perspective. OR Spectrum, 43, 1–58.
- Dixon, M., Freeman, K., & Toman, N. (2020). Digital POS systems and their impact on restaurant operations. Cornell Hospitality Quarterly, 61(2), 45–60.
- Google Developers. (2023). Flutter: Build apps for any screen. https://flutter.dev
- Gupta, S. (2016). The impact of technology on restaurant operations. International Journal of Hospitality Management, 55, 91–102.
- Heizer, J., Render, B., & Munson, C. (2019). Operations Management: Sustainability and Supply Chain Management (13th ed.). Pearson.
- Kasavana, M. L., & Cahill, J. J. (2007). Managing Computers in the Hospitality Industry (5th ed.). American Hotel & Lodging Educational Institute.
- Law, R., Buhalis, D., & Cobanoglu, C. (2013). Progress on information and communication technologies in hospitality and tourism. International Journal of Contemporary Hospitality Management, 26(5), 727–750.
- National Restaurant Association. (2023). Restaurant Technology Landscape Report 2023. https://restaurant.org
- Pigatto, G., Machado, J. G. C. F., Negreti, A. S., & Machado, L. M. (2017). Have you chosen your request? Analysis of online food delivery companies in Brazil. British Food Journal, 119(3), 639–657.
- Statista. (2024). Online food delivery market worldwide. https://www.statista.com

1.4 Overview of Document

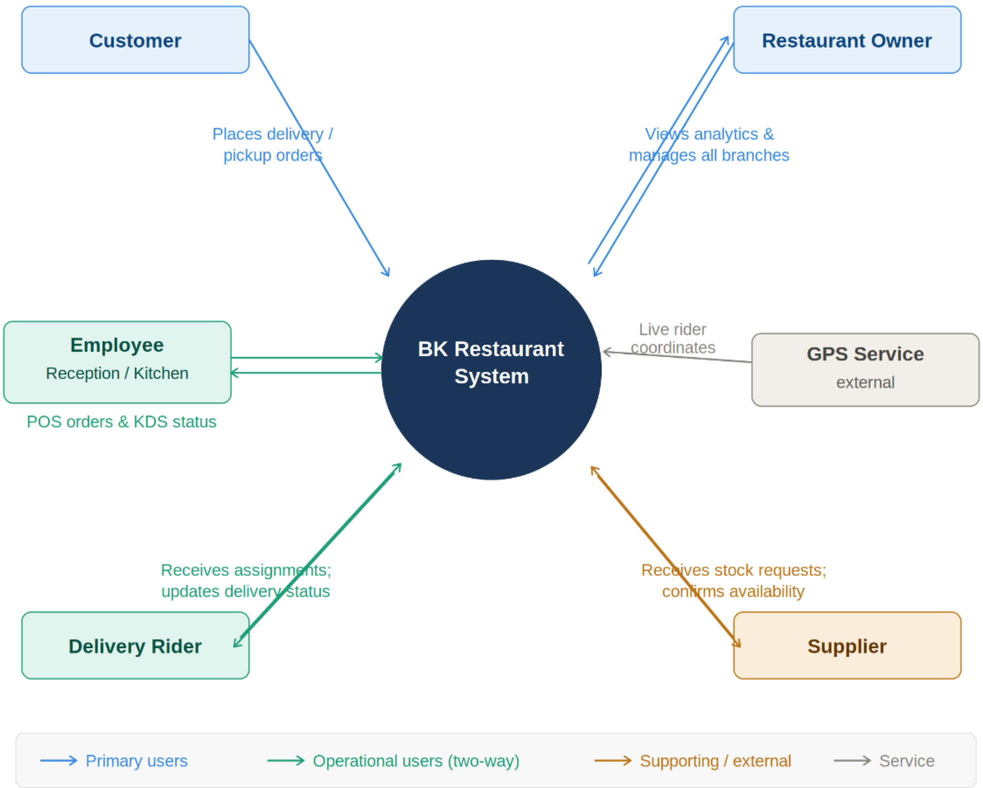
This document follows the standard SRS structure for SENG 31242:

1. Section 1 – Introduction: purpose, scope, definitions, acronyms, references.
2. Section 2 – Overall Description: product perspective, user classes, operating environment, constraints, and assumptions.
3. Section 3 – System Analysis: fact-gathering techniques, existing system analysis, use-case diagrams, use-case descriptions, activity diagrams, and system sequence diagrams.
4. Section 4 – Functional Requirements: numbered FR list (FR-01, FR-02, …) with ID, title, description, priority, source, and verification method.
5. Section 5 – Non-Functional Requirements: performance, security, usability, reliability, scalability, and maintainability.
6. Section 6 – Alternative Solutions & Feasibility Study: at least two alternatives analysed, four-dimension feasibility assessment, and justified recommendation.

2. Overall Description

2.1 Product Perspective (Context Diagram)

The BK Restaurant Management System is a purpose-built, integrated platform replacing the existing fragmented manual operations. The system boundary encompasses three primary sub-systems: a web-based POS dashboard, a Flutter mobile application for riders, and a centralized backend API. The diagram below shows the system context and all external entities.



External Entity	Interaction	Direction
Customer	Places external orders; receives order confirmation	Input
Restaurant Owner	Monitors all branches; views analytics; manages staff and inventory	Both
Employee (Reception/Kitchen)	Processes walk-in orders via POS; updates kitchen status	Both
Delivery Rider	Receives delivery assignments on mobile app; updates delivery status	Both
Supplier	Receives automated stock requests; confirms availability	Both
GPS Service (ext.)	Provides real-time location data for rider tracking	Input

2.2 User Classes and Characteristics (Personas)

User Class	Persona	Tech Level	Primary Interface	Key Responsibilities
Restaurant Owner	Mr. Sanjeewa, 45, business owner across 3 branches. Comfortable with web apps.	Medium	Web Dashboard	Monitor branches, view financials, manage inventory, approve schedules
Reception Staff	Amali, 28, counter cashier. Limited tech experience.	Low–Medium	POS System	Create orders, generate bills, manage walk-ins
Kitchen Staff	Nishantha, 32, cook. Uses only the KDS screen.	Low	KDS Display	View and update order preparation status
Delivery Rider	Kusal, 24, motorcycle rider. Smartphone-literate.	Low	Mobile App	Receive assignments, navigate, update delivery status
Supplier	External vendor. Interacts via email/SMS notification.	Varies	Notification	Receive stock requests, confirm availability

2.3 Operating Environment

Component	Operating Environment
POS and Management System	Web-based application accessible through desktop computers, laptops, and tablets used by restaurant staff and management.
Client Platform	Modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
Frontend Environment	React.js with Tailwind CSS for responsive and user-friendly interfaces.
Backend Environment	Node.js and Express.js running on a cloud-hosted server to support system functionality and API services.
Database Environment	PostgreSQL/MySQL database for storing orders, inventory, employee, customer, and financial data.
Hardware Environment	Desktop computers, tablets, POS terminals, receipt printers, and smartphones used for restaurant operations and delivery activities.
Network Environment	Internet connectivity is required for real-time synchronization between branches, online ordering, reporting, and external service integrations.
Hosting Environment	Cloud-based hosting platform such as AWS or Railway.app to provide system availability and scalability.

2.4 Design and Implementation Constraints

- System must function effectively in areas with poor internet connectivity (Kahawa/Ambalangoda), requiring local caching and synchronization mechanisms.
- Must remain cost-effective to avoid the high overhead of global SaaS competitors.
- UI must be extremely intuitive for non-technical staff, requiring minimal training.
- System shall be developed using a modular architecture (React.js frontend and Node.js backend) to support future enhancements and maintenance.
- System must support secure handling of customer, order, and business data through authentication and encryption mechanisms.
- System shall maintain reliable performance during peak restaurant operating hours when multiple users access the system simultaneously.

2.5 Assumptions and Dependencies

- Delivery riders possess personal smartphones with functioning GPS and internet connectivity for delivery tracking and navigation (if delivery module is enabled).
- Branches have the necessary hardware (PCs/Tablets and receipt printers) to support POS operations and order processing.
- A stable internet connection is available at restaurant branches to ensure real-time synchronization of orders, billing, and inventory data.
- Restaurant staff and management have basic digital literacy to operate the system with minimal training.
- The system depends on third-party services such as payment gateways, SMS/email notification services, and mapping APIs (if delivery tracking is included) for full functionality.

3. System Analysis

3.1 Fact-Gathering Techniques Used

The following methods were used to collect, understand, and confirm the requirements for the system:

Technique	Description	Output / Evidence Produced	Purpose in the Project
Structured Client Interviews	Conducted structured interviews with the client (Mr. N. C. Sanjeewa De Silva) using prepared questions to understand system requirements, business workflow, and pain points.	Interview transcript document and summarized notes, zoom meeting recording	To gather direct and accurate user requirements and real-world operational needs of the restaurant system
Competitor Analysis	Analysed existing restaurant management systems and compared their features, strengths, and weaknesses.	Feature comparison table	To identify industry standards, missing features, and improvement opportunities for the proposed system
Literature Review/ Document Analysis	Reviewed academic papers and industry sources related to restaurant management systems, POS systems, and digital ordering systems.	Literature review document	To understand existing technologies, best practices, and support system design decisions

Evidence Summary:

Evidence was gathered through a structured interview conducted with the owner of BK Restaurant to understand the challenges faced in managing operations across multiple branches. The interview revealed that restaurant activities, including order processing, inventory management, delivery coordination, and employee administration, are currently handled manually using WhatsApp messages, phone calls, and paper-based records. This has resulted in issues such as missed customer orders, lack of real-time delivery visibility, inconsistent inventory tracking, and limited oversight of employee attendance and schedules.

Key findings highlighted the need for a centralized management solution that would enable the owner to monitor all branches from a single dashboard, digitize order processing, track deliveries in real time, automate inventory management, and streamline employee-related activities. The client also emphasized the importance of automated alerts, supplier request management, and comprehensive reporting capabilities.

The findings were supported by the following evidence artifacts maintained within the project repository:

1. Client Interview Questions
2. Client Interview Session notes
3. Structured Interview Transcript Documentation
4. Competitor Analysis (Feature Comparison Table)
5. Literature Review (Academic and Industry Sources)

These artifacts provided the foundation for identifying business requirements, validating stakeholder needs, and defining the functional and non-functional requirements of the proposed BK Restaurant Management System.

3.2 Existing System Analysis (with Pain Points)

The current restaurant management process is largely manual and semi-digital, relying on WhatsApp messages, phone calls, physical registers, and spreadsheets. Each branch operates with limited integration, meaning data is not shared in real time across locations. This leads to inefficiencies in order handling, inventory tracking, and financial reporting.

3.2.1 Current System Overview

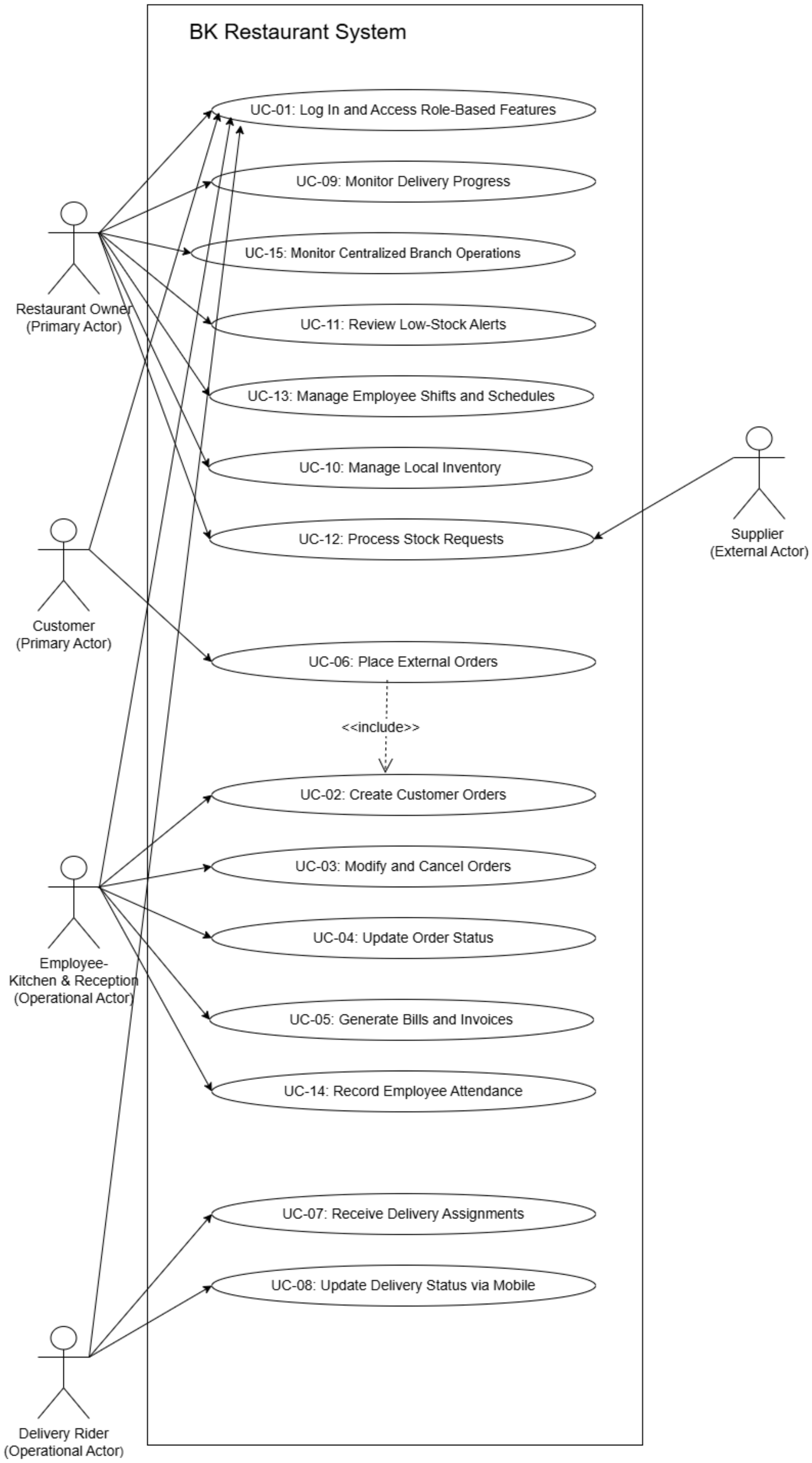
The existing system at the restaurant primarily uses WhatsApp for order communication between customers and staff, while kitchen operations and billing are managed through manual notes or basic POS tools. Delivery operations are handled without real-time tracking, and riders are dispatched based on availability without system assistance. Inventory and supplier management are done manually through physical logs and phone calls. Financial reporting across multiple branches is done at the end of each day using spreadsheets or paper records.

3.2.2 Pain Points Identified

Pain Point	Description	Impact on Business	Proposed Solution (Our System)
Missed or duplicated orders	Orders received via WhatsApp are manually tracked, leading to confusion during busy hours	Customer dissatisfaction and loss of revenue	Centralized order management system where all orders are placed and tracked in real time through a unified POS dashboard
Lack of delivery visibility	Delivery riders are dispatched without real-time tracking or status updates	Delays in delivery and poor customer experience	Real-time delivery tracking system with GPS-enabled rider updates and live order status monitoring
Inventory stockouts	Stock levels are updated manually, leading to delayed awareness of low inventory	Frequent stockouts and disruption in service	Automated inventory management system with low-stock alerts and real-time stock updates across branches
Manual financial reconciliation	Cross-branch financial reports are prepared manually using spreadsheets	Time-consuming and error-prone financial analysis	Automated financial reporting system with real-time dashboards and cross-branch analytics
Paper-based staff management	Staff schedules and shifts are managed using paper or informal communication	Scheduling conflicts and inefficient workforce allocation	Digital staff management and scheduling module with centralized shift planning and updates

3.3 Use Case Diagram (System Boundary)

The use case diagram below illustrates the system boundary and the interactions between all actors and the BK Restaurant Management System. The system covers five primary actors: Restaurant Owner, Customer, Employee (Reception/Kitchen), Delivery Rider, and Supplier.



3.4 Use Case Descriptions (one per use case)

The following use cases define the complete scope of system interactions. Full detailed descriptions for each use case are provided below.

UC-01: Log In and Access Role-Based Features

Use Case ID	UC-01
Use Case Name	Log In and Access Role-Based Features
Actor(s)	Restaurant Owner, Branch Managers, Employees/Cashiers, Delivery Riders, Suppliers
Preconditions	User is registered in the system and is not currently logged in.
Postconditions – Success	User is authenticated and redirected to their role-specific dashboard.
Postconditions – Failure	Authentication fails; user remains on the login page.
Main Flow	1. User navigates to the login screen. 2. User enters username and password. 3. System validates credentials. 4. System retrieves user role. 5. System grants access to role-based features.
Alternative Flow 1	At step 3: If credentials are invalid, system displays an 'Invalid login' error message.
Alternative Flow 2	At step 2: User clicks 'Forgot Password', redirecting to password reset flow.
Business Rules	Users can only access modules permitted by their assigned role.

UC-02: Create Customer Orders

Use Case ID	UC-02
Use Case Name	Create Customer Orders
Actor(s)	Employees/Cashiers
Preconditions	Employee is logged in to the POS module.
Postconditions – Success	Order is recorded, inventory is marked for update, and kitchen is notified.
Postconditions – Failure	Order is discarded; no inventory changes occur.
Main Flow	1. Employee initiates new order. 2. Employee selects order type (Dine-in, Takeaway, Delivery). 3. Employee adds menu items. 4. System calculates total. 5. Employee confirms order and accepts payment. 6. System saves order.
Alternative Flow 1	At step 3: If an item is out of stock, system prevents adding it and displays an alert.
Alternative Flow 2	None
Business Rules	Order total must be greater than zero.

UC-03: Modify and Cancel Orders

Use Case ID	UC-03
Use Case Name	Modify and Cancel Orders
Actor(s)	Employees/Cashiers
Preconditions	Employee is logged in; an active order exists.
Postconditions – Success	Order is successfully updated or cancelled; system logs the change.
Postconditions – Failure	Order remains in its previous state.
Main Flow	1. Employee searches for an active order. 2. System displays order details. 3. Employee selects 'Modify' or 'Cancel'. 4. Employee makes changes or confirms cancellation. 5. System updates order status and recalculates totals if modified.
Alternative Flow 1	At step 3: If order is already in 'Dispatched' or 'Completed' status, system denies modification.
Alternative Flow 2	None
Business Rules	Orders cannot be cancelled once they are dispatched.

UC-04: Update Order Status

Use Case ID	UC-04
Use Case Name	Update Order Status
Actor(s)	Employees/Cashiers
Preconditions	Employee is logged in; active orders exist in the queue.
Postconditions – Success	Order status is updated (e.g., Preparing, Ready, Completed).
Postconditions – Failure	Status update fails; error logged.
Main Flow	1. Employee views active order queue. 2. Employee selects an order. 3. Employee changes status from 'Pending' to 'Preparing'. 4. System saves the new status. 5. Status updates on real-time displays.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Status progression must follow a logical sequence (Pending -> Preparing -> Ready).

UC-05: Generate Bills and Invoices

Use Case ID	UC-05
Use Case Name	Generate Bills and Invoices
Actor(s)	Employees/Cashiers
Preconditions	Order has been created and confirmed.
Postconditions – Success	Digital bill/invoice is generated and stored in order history.
Postconditions – Failure	Invoice generation fails; user is prompted to retry.
Main Flow	1. Employee finalizes an order. 2. System calculates final amount including taxes/fees. 3. System generates a digital invoice. 4. System offers option to print or email the invoice. 5. System archives invoice in order history.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Invoices must include a unique transaction ID and timestamp.

UC-06: Place External Orders

Use Case ID	UC-06
Use Case Name	Place External Orders
Actor(s)	Customers
Preconditions	Customer has access to phone or WhatsApp.
Postconditions – Success	Order is received by the restaurant and entered into the system.
Postconditions – Failure	Order is not placed (e.g., outside delivery hours).
Main Flow	1. Customer contacts restaurant via Phone/WhatsApp. 2. Customer provides order details and delivery address. 3. Employee enters details into the system (invoking UC-02). 4. Customer receives confirmation.
Alternative Flow 1	At step 2: If address is outside delivery zone, customer is informed.
Alternative Flow 2	None
Business Rules	Delivery orders require a valid contact number and address.

UC-07: Receive Delivery Assignments

Use Case ID	UC-07
Use Case Name	Receive Delivery Assignments
Actor(s)	Delivery Riders
Preconditions	Rider is logged into the mobile app and marked as 'Available'.
Postconditions – Success	Order is assigned to the rider; order status becomes 'Dispatched'.
Postconditions – Failure	Assignment fails; order remains 'Ready for Delivery'.
Main Flow	1. System identifies a 'Ready' delivery order. 2. System assigns order to an available rider. 3. Rider receives notification on mobile app. 4. Rider accepts assignment. 5. System updates order status to 'Dispatched'.
Alternative Flow 1	At step 4: Rider rejects assignment; system reassigns to the next available rider.
Alternative Flow 2	None
Business Rules	Riders can only be assigned orders from their designated branch.

UC-08: Update Delivery Status via Mobile

Use Case ID	UC-08
Use Case Name	Update Delivery Status via Mobile
Actor(s)	Delivery Riders
Preconditions	Rider has an active assigned delivery.
Postconditions – Success	Delivery is marked 'Completed'; customer notified.
Postconditions – Failure	Status remains unchanged; rider prompted to retry.
Main Flow	1. Rider arrives at delivery location. 2. Rider hands over the order. 3. Rider opens mobile app. 4. Rider marks order as 'Delivered'. 5. System logs completion time and updates central database.
Alternative Flow 1	At step 2: Customer is unavailable; rider marks order as 'Delivery Failed'.
Alternative Flow 2	None
Business Rules	Riders must be within reasonable GPS proximity to the delivery address to mark as delivered.

UC-09: Monitor Delivery Progress

Use Case ID	UC-09
Use Case Name	Monitor Delivery Progress
Actor(s)	Restaurant Owner
Preconditions	Owner is logged in; deliveries are currently active.
Postconditions – Success	Owner successfully views real-time delivery locations.
Postconditions – Failure	Map or tracking data fails to load.
Main Flow	1. Owner navigates to 'Delivery Tracking' dashboard. 2. System retrieves GPS coordinates of active riders. 3. System overlays rider locations and estimated delivery times on a map view. 4. Owner selects a specific rider for detailed order info.
Alternative Flow 1	At step 2: If a rider's GPS is off, system displays their last known location and a warning.
Alternative Flow 2	None
Business Rules	GPS data must be updated at least every 60 seconds.

UC-10: Manage Local Inventory

Use Case ID	UC-10
Use Case Name	Manage Local Inventory
Actor(s)	Branch Managers
Preconditions	Branch Manager is logged into the inventory module.
Postconditions – Success	Inventory counts are accurately adjusted and saved.
Postconditions – Failure	Inventory update fails; previous counts remain.
Main Flow	1. Manager navigates to 'Inventory Management'. 2. System displays current stock levels. 3. Manager performs manual stock count. 4. Manager enters adjustments into the system. 5. System saves updated inventory records.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Inventory updates automatically after sales; manual adjustments require a logged reason.

UC-11: Review Low-Stock Alerts

Use Case ID	UC-11
Use Case Name	Review Low-Stock Alerts
Actor(s)	Branch Managers
Preconditions	Inventory item falls below the defined threshold.
Postconditions – Success	Manager acknowledges alert and initiates a stock request.
Postconditions – Failure	Manager ignores alert; item risks stocking out.
Main Flow	1. System detects an item count below minimum threshold. 2. System generates a low-stock alert. 3. Manager reviews alerts on dashboard. 4. Manager acknowledges the alert. 5. Manager proceeds to create a stock request.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Alerts remain active until inventory count is increased above the threshold.

UC-12: Process Stock Requests

Use Case ID	UC-12
Use Case Name	Process Stock Requests
Actor(s)	Branch Managers, Suppliers
Preconditions	Manager has identified a need for supplies.
Postconditions – Success	Stock request is submitted and fulfilled by supplier.
Postconditions – Failure	Stock request is rejected or fails to send.
Main Flow	1. Manager creates a new stock request. 2. Manager adds items and quantities. 3. System sends request to corresponding Supplier. 4. Supplier logs in and reviews request. 5. Supplier approves and dispatches items. 6. Manager confirms receipt.
Alternative Flow 1	At step 5: Supplier lacks stock and rejects/modifies the request.
Alternative Flow 2	None
Business Rules	Received stock automatically updates branch inventory counts.

UC-13: Manage Employee Shifts and Schedules

Use Case ID	UC-13
Use Case Name	Manage Employee Shifts and Schedules
Actor(s)	Branch Managers
Preconditions	Manager is logged into the employee module.
Postconditions – Success	Weekly schedule is published and accessible to employees.
Postconditions – Failure	Schedule is not saved.
Main Flow	1. Manager opens the scheduling tool. 2. Manager assigns employees to specific shifts. 3. System checks for scheduling conflicts. 4. Manager finalizes the schedule. 5. System notifies employees of their upcoming shifts.
Alternative Flow 1	At step 3: System detects an overlapping shift and prompts manager to resolve it.
Alternative Flow 2	None
Business Rules	Employees cannot be scheduled for overlapping shifts.

UC-14: Record Employee Attendance

Use Case ID	UC-14
Use Case Name	Record Employee Attendance
Actor(s)	Employees/Cashiers
Preconditions	Employee is physically at the branch and has system access.
Postconditions – Success	Clock-in/clock-out time is recorded in the system.
Postconditions – Failure	Attendance record fails to save.
Main Flow	1. Employee accesses the attendance module. 2. Employee enters unique pin or credentials. 3. Employee selects 'Clock In' (or 'Clock Out'). 4. System logs the timestamp and employee ID. 5. System displays confirmation.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Clock-in must occur within the designated branch network or location.

UC-15: Monitor Centralized Branch Operations & Analytics

Use Case ID	UC-15
Use Case Name	Monitor Centralized Branch Operations & Analytics
Actor(s)	Restaurant Owner
Preconditions	Owner is logged into the central admin dashboard.
Postconditions – Success	Owner successfully generates and views performance reports.
Postconditions – Failure	Data fails to aggregate; dashboard displays an error.
Main Flow	1. Owner opens the analytics dashboard. 2. System aggregates data from all branches. 3. System displays sales, inventory, and delivery metrics. 4. Owner applies filters (e.g., date range, specific branch). 5. System updates visual charts and data tables.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Analytics dashboard is strictly restricted to the Restaurant Owner role.

3.5 Activity Diagrams (one per major use case)

The following activity diagrams illustrate the detailed workflows of the major use cases in the BK Restaurant Management System. Each diagram shows the interactions between actors and the system, including decision points, validations, notifications, and exception handling scenarios.

AD-01: User Login and Role-Based Access

Description:
This activity diagram illustrates the authentication process of the BK Restaurant Management System. The user enters login credentials, and the system validates them. If the credentials are valid, the system retrieves the user's role and redirects the user to the appropriate dashboard with role-based features. Invalid credentials result in access denial and a request to re-enter credentials.

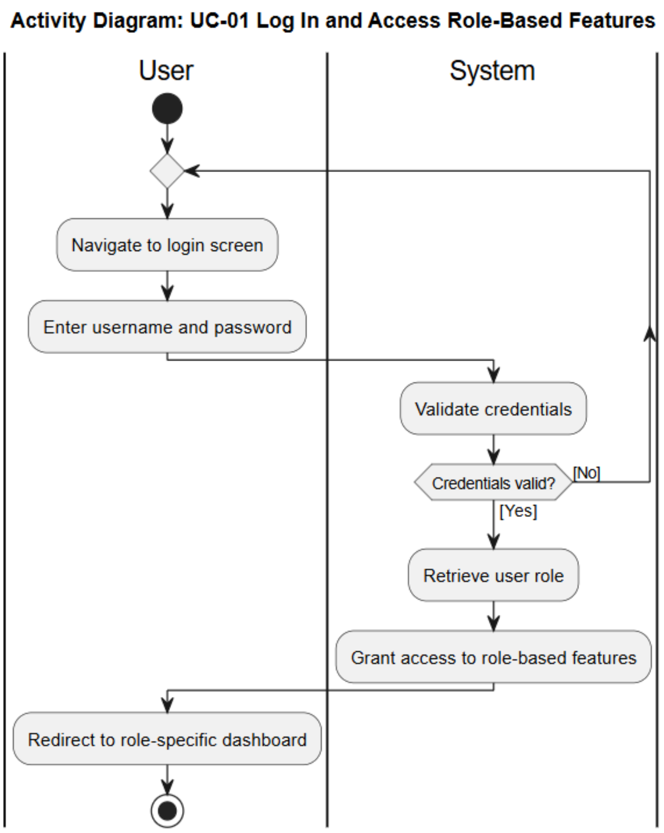
Actors Involved:

- User
- System

Use Cases Covered:

- UC-01 Log In and Access Role-Based Features

Activity Diagram:



AD-02: Update Order Status

Description:
This activity diagram illustrates how kitchen staff manage the order preparation workflow. The kitchen selects pending orders, updates their status to “Preparing,” completes food preparation, and marks orders as “Ready.” The system automatically notifies the cashier or delivery rider. If issues such as missing ingredients occur, the system records the issue and allows cancellation or refund procedures.

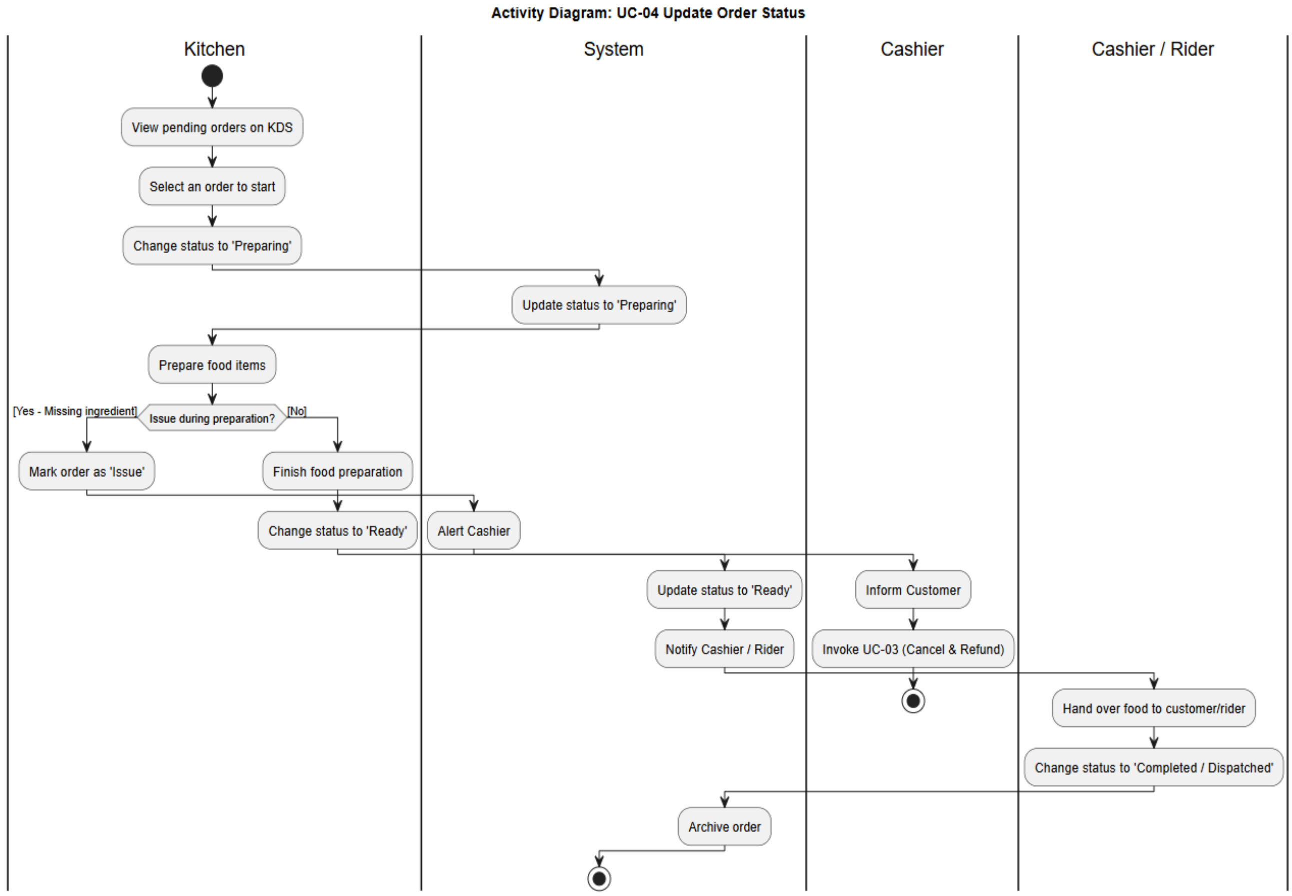
Actors Involved:

- Kitchen Staff
- System
- Cashier
- Rider

Use Cases Covered:

- UC-04 Update Order Status

Activity Diagram:



AD-03: Generate Bills and Invoices

Description:
This activity diagram describes the invoice generation process after an order is finalized. The system calculates taxes and fees, generates a digital invoice, and allows the cashier to either print or email the invoice. Error handling mechanisms are included for invoice generation failures and printer connectivity issues.

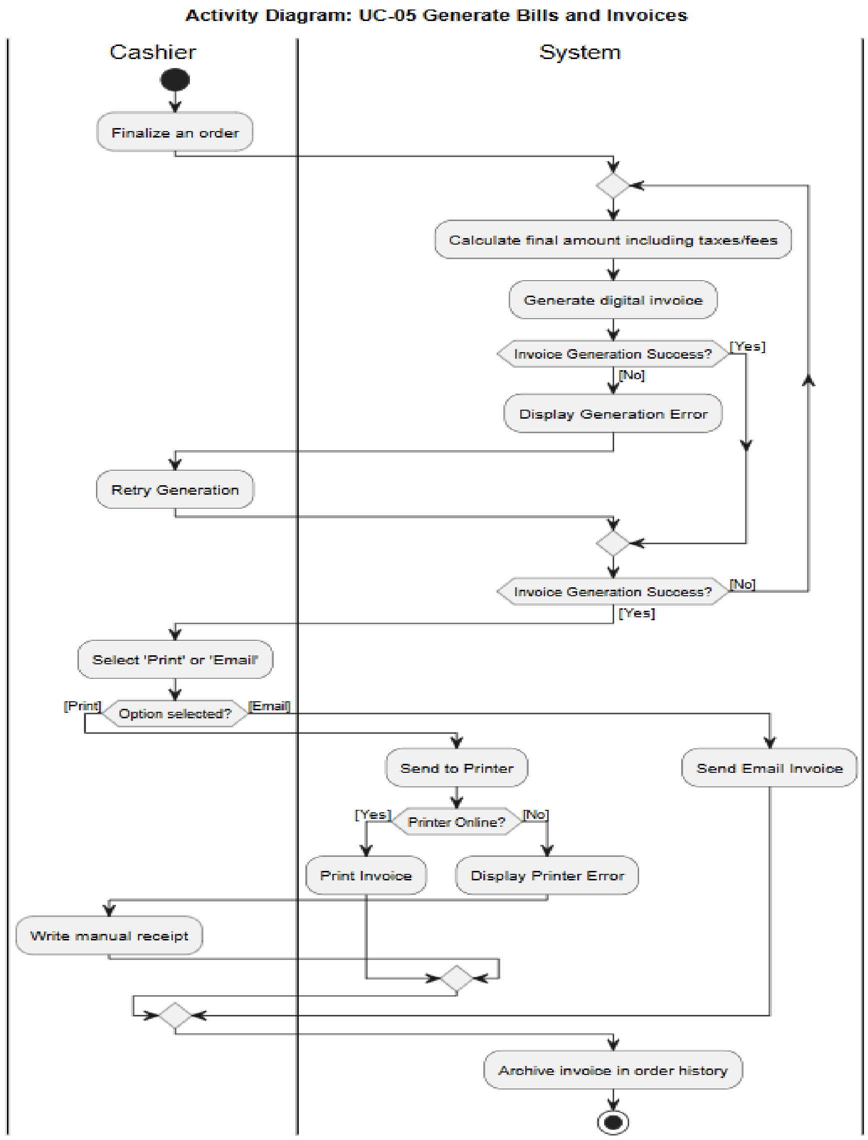
Actors Involved:

- Cashier
- System

Use Cases Covered:

- UC-05 Generate Bills and Invoices

Activity Diagram:



AD-04: Receive Delivery Assignments

Use Case: UC-07 Receive Delivery Assignments

Description:
This activity diagram illustrates the delivery assignment process. The system identifies a ready order and checks rider availability. Available riders receive delivery notifications through the mobile application and may accept or reject assignments. If no riders are available, the system generates alerts and supports alternative delivery handling procedures.

Actors Involved:

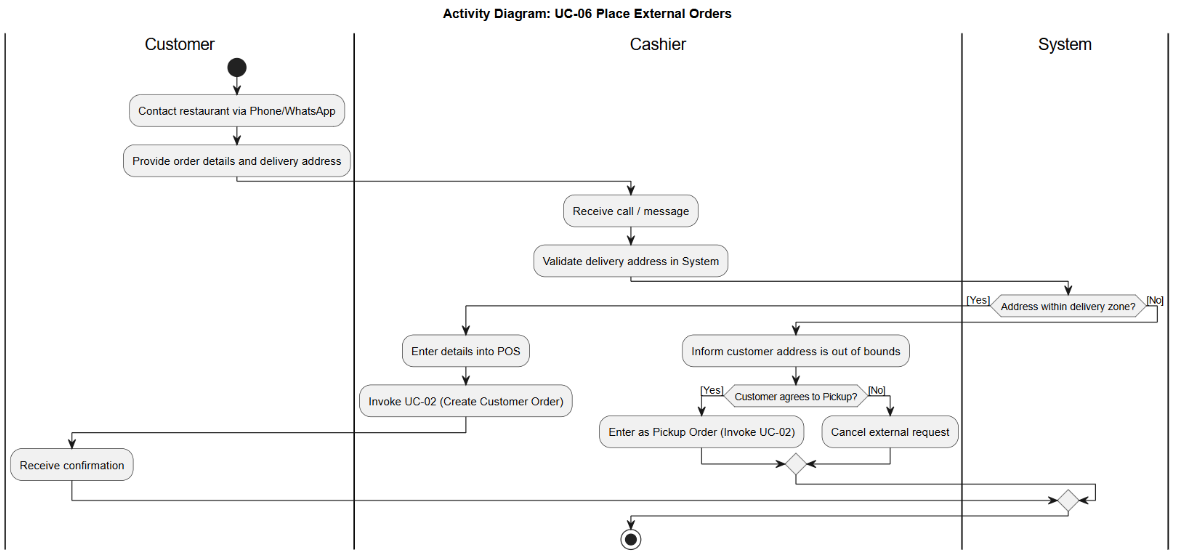
- System
- Delivery Rider
- Branch Manager

Use Cases Covered:

- UC-07 Receive Delivery Assignments

Activity Diagram:

Figure 3.5.4: Activity Diagram for Receive Delivery Assignments (UC-07)



AD-05: Monitor Delivery Progress

Description:
This activity diagram illustrates how restaurant owners monitor ongoing deliveries. The system continuously tracks rider GPS locations and updates estimated delivery times. If GPS tracking becomes unavailable, the system displays warnings and the rider's last known location to support delivery supervision.

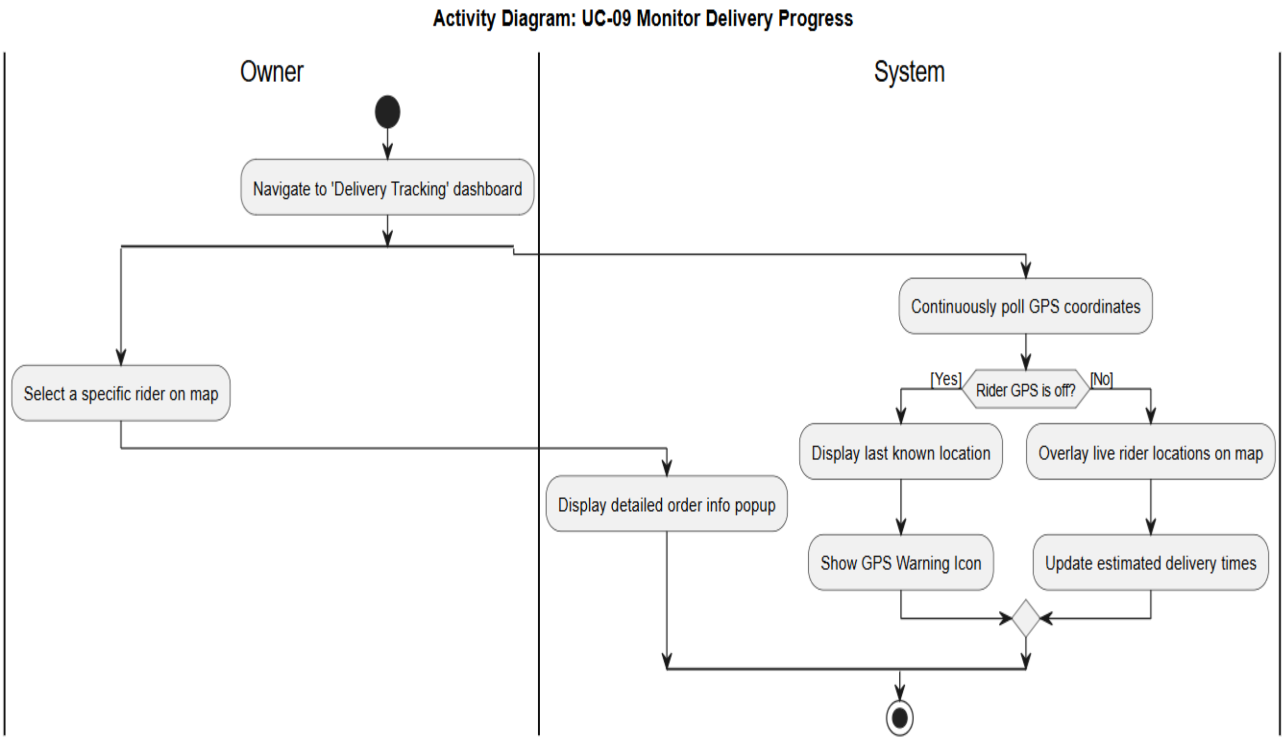
Actors Involved:

- Restaurant Owner
- System

Use Cases Covered:

- UC-09 Monitor Delivery Progress

Activity Diagram:



AD-06: Manage Local Inventory

Description:
This activity diagram illustrates inventory management activities performed by branch managers. Managers review inventory levels, perform physical stock counts, compare counts with system records, and make stock adjustments when discrepancies are identified. The system records adjustment reasons, timestamps, and manager information for auditing purposes.

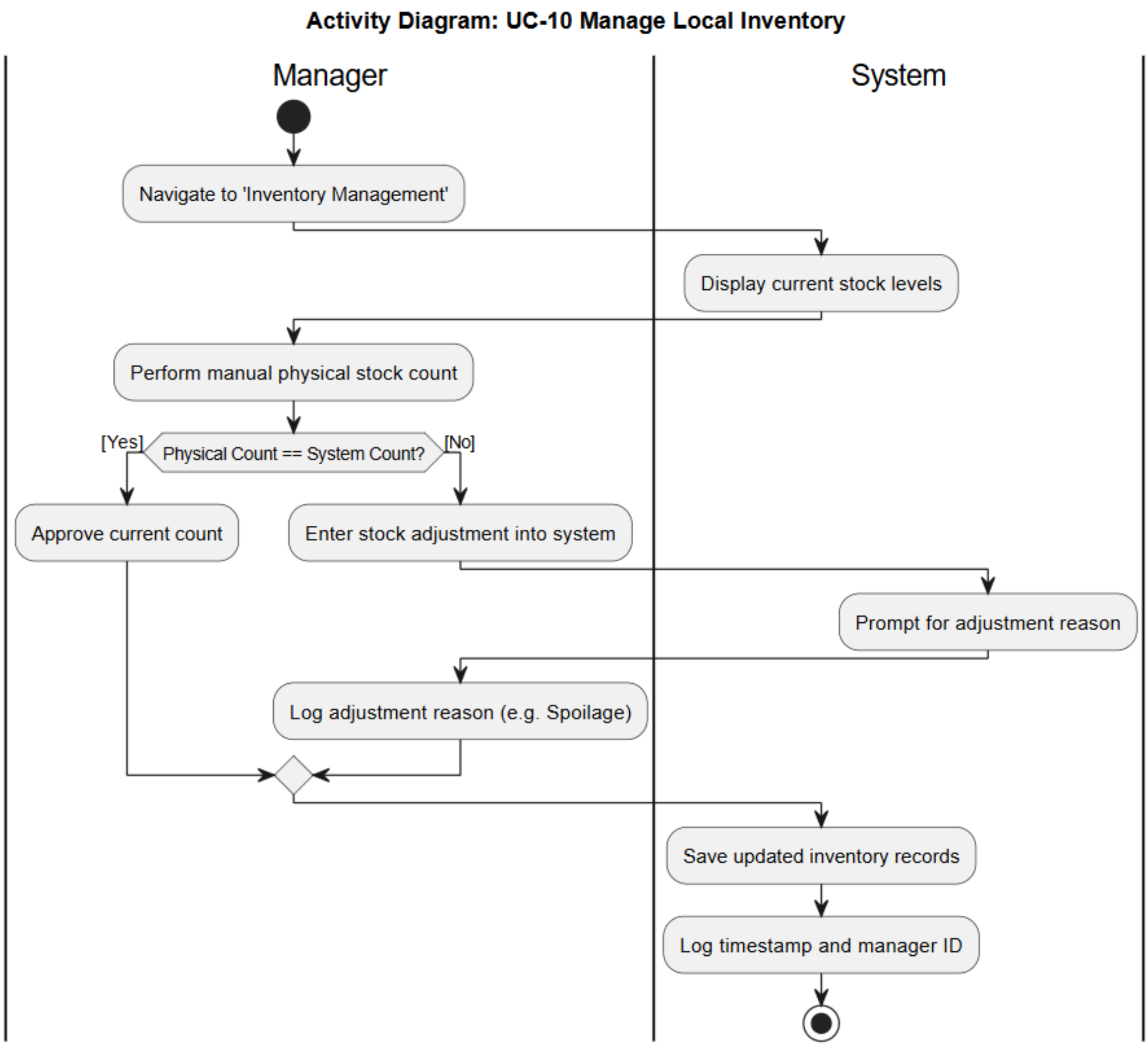
Actors Involved:

- Branch Manager
- System

Use Cases Covered:

- UC-10 Manage Local Inventory

Activity Diagram:

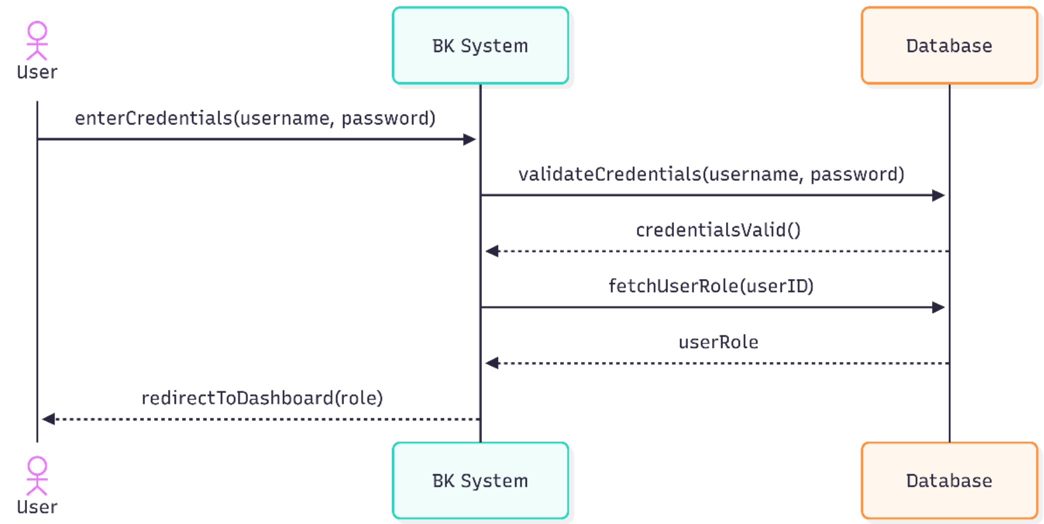


3.6 System Sequence Diagrams (High-Level)

System Sequence Diagrams show the high-level interactions between actors and the BK Restaurant System boundary for the key use cases.

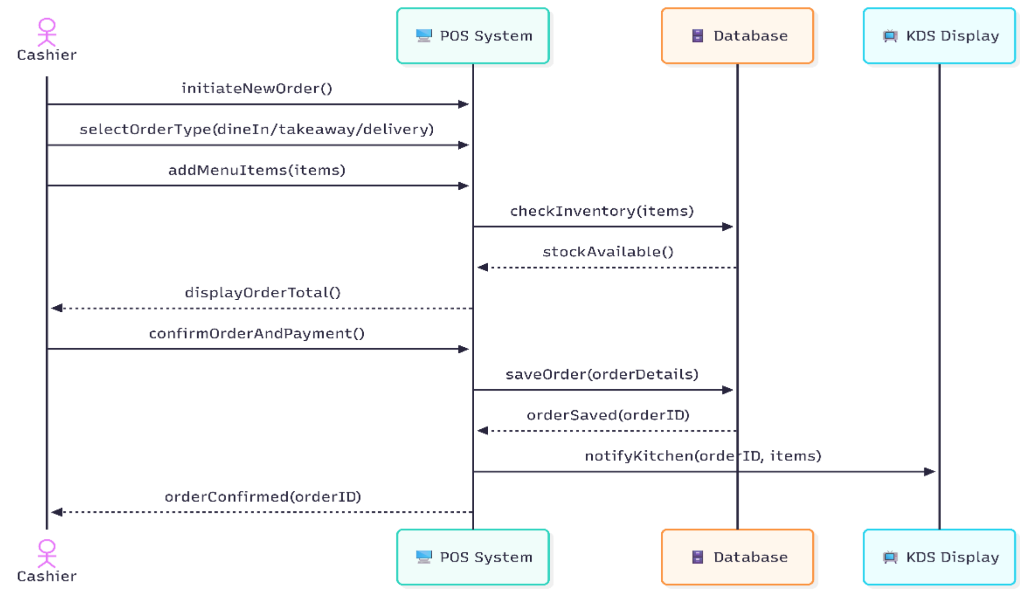
SSD-01 — Log In and Access Role-Based Features

This diagram shows the user authentication process in the BK Restaurant Management System. After login, the system verifies the user's credentials, identifies their role, and redirects them to the appropriate dashboard with role-based access and permissions.



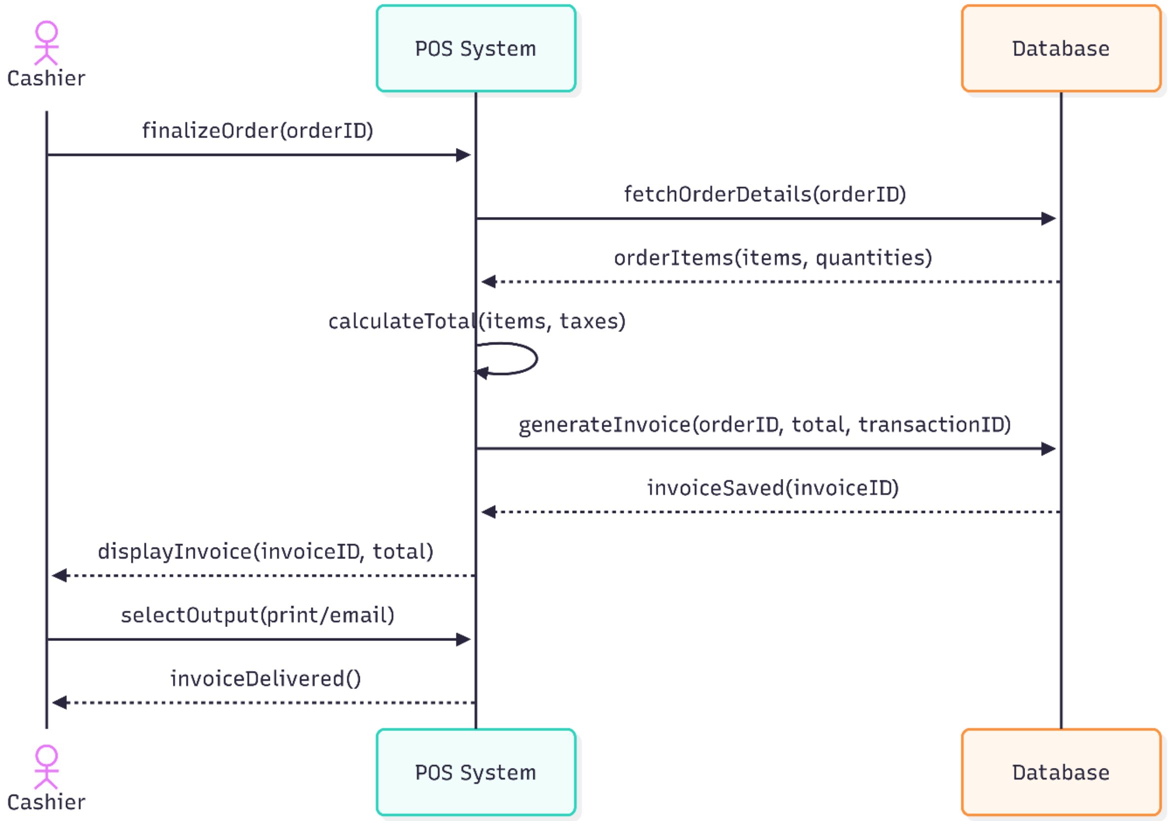
SSD-02 — Create Customer Orders

This diagram illustrates the order creation process in the BK Restaurant Management System. The Cashier selects the order type, adds menu items, and confirms the order after payment. The system verifies inventory availability, saves the order, and instantly notifies the Kitchen Display System (KDS) for preparation.



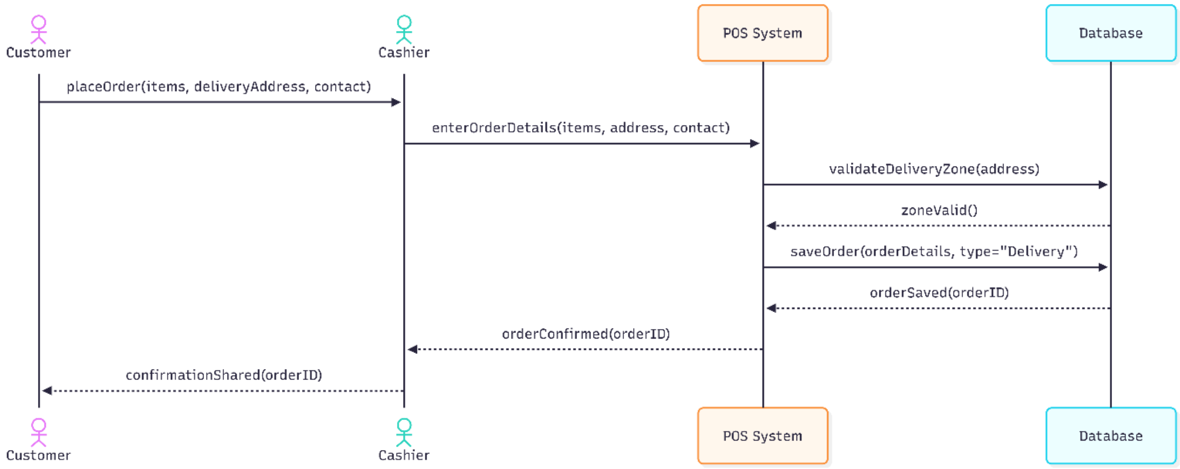
SSD-03 — Generate Bills and Invoices

This diagram illustrates the billing process in the BK Restaurant Management System. The Cashier finalizes the order, the system calculates the total amount and generates an invoice, then saves it with a unique reference. The invoice is delivered to the customer either as a printed copy or via email, completing the transaction.



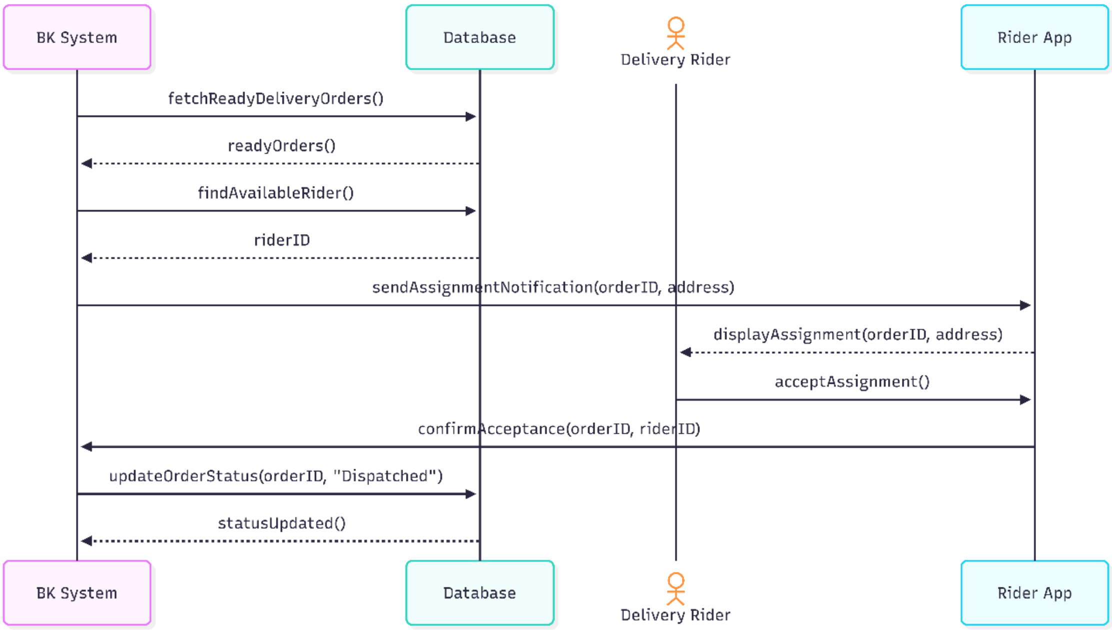
SSD-04 — Place External Orders

This diagram illustrates the delivery order process in the BK Restaurant Management System. The Cashier records the customer's order and delivery details, and the system verifies that the address is within the delivery zone. Once validated, the order is saved as a delivery order, and the customer receives an order ID for tracking purposes.



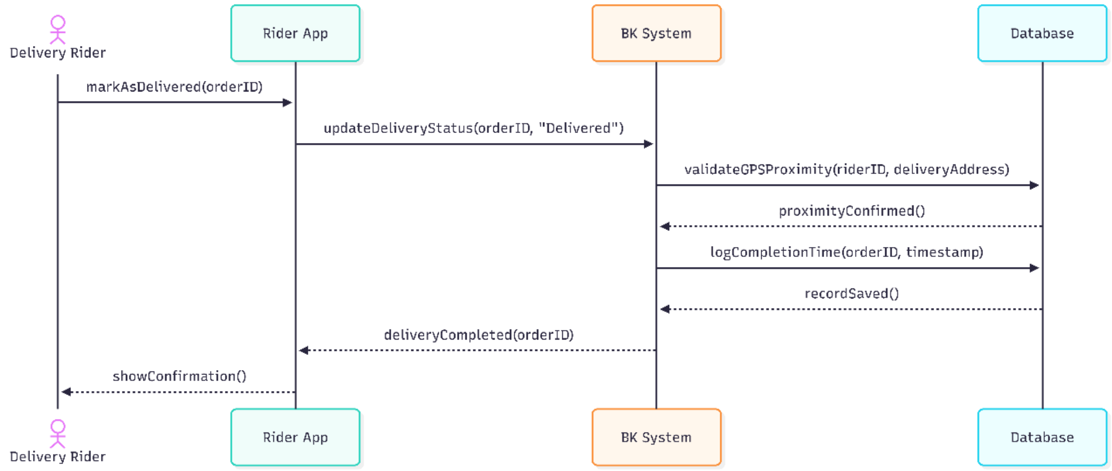
SSD-05 — Receive Delivery Assignments

This diagram illustrates the rider assignment process in the BK Restaurant Management System. When an order is marked as ready, the system automatically assigns it to an available rider and sends the delivery details. After the rider accepts the assignment, the system updates the order status to "Dispatched," initiating the delivery process.



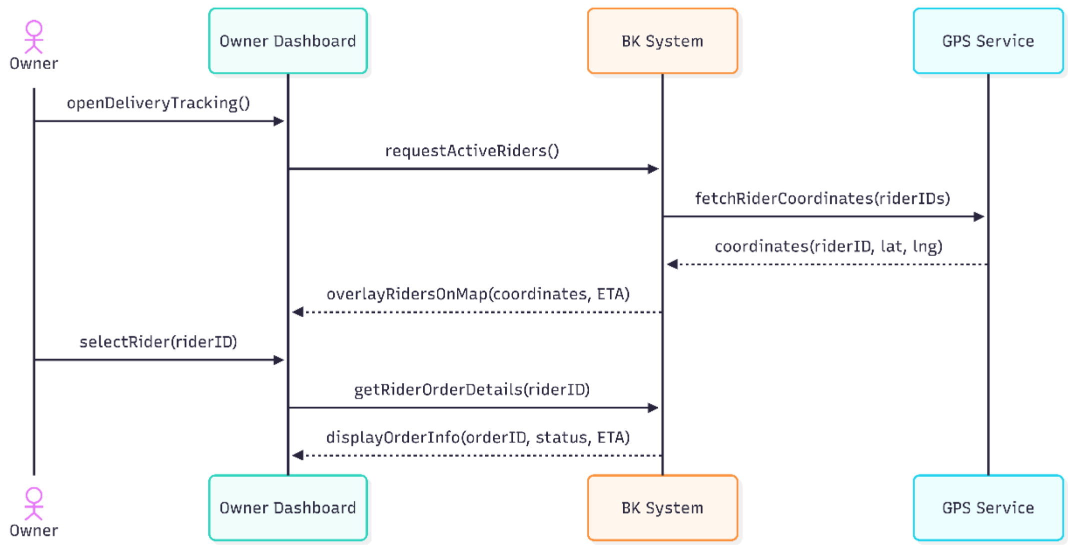
SSD-06 — Update Delivery Status via Mobile

This diagram illustrates the delivery completion process in the BK Restaurant Management System. After delivering an order, the rider updates its status to "Delivered" through the mobile app. The system verifies the rider's location, records the delivery completion time, and confirms the successful update, completing the delivery process.



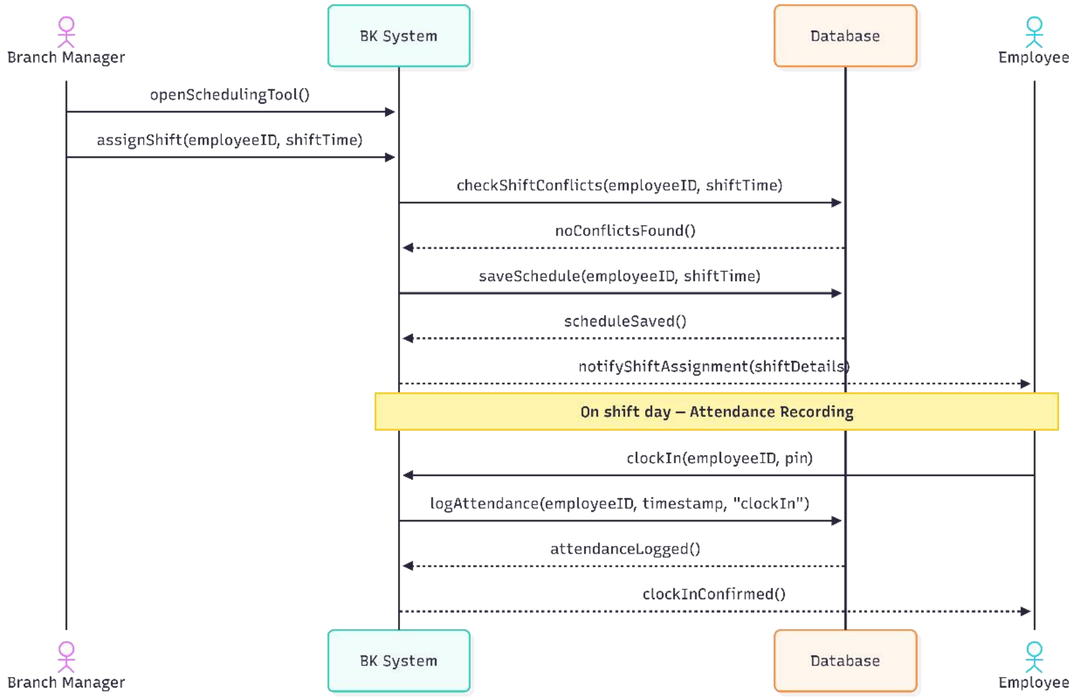
SSD-07 — Monitor Delivery Progress

This diagram shows the Owner's real-time delivery tracking dashboard. It pulls live GPS data to display rider locations on a map with ETAs, and allows the Owner to view detailed information for each rider and order, ensuring full visibility of all active deliveries.



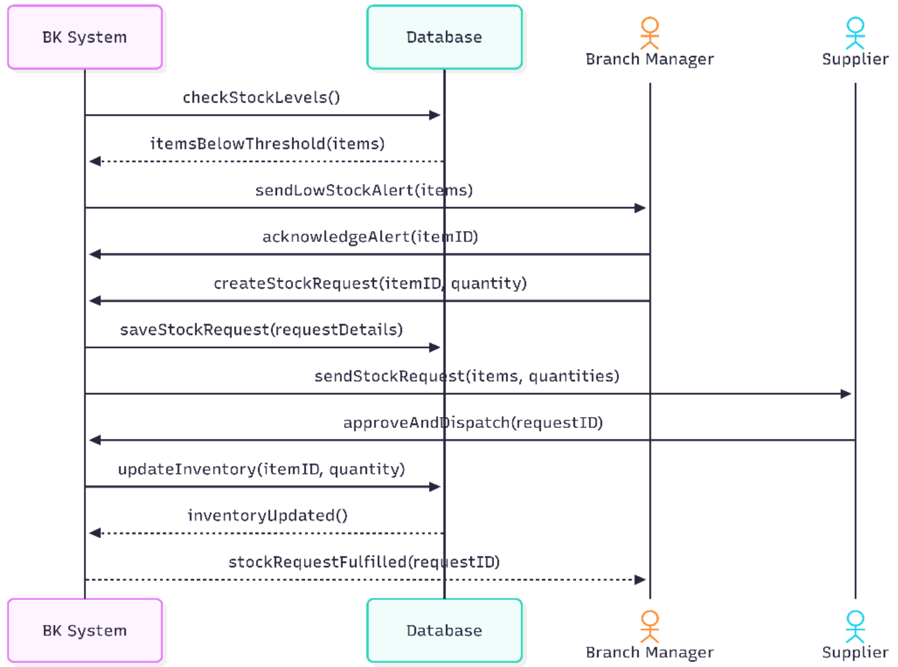
SSD-08 — Manage Employee Shifts and Record Attendance

This diagram illustrates shift scheduling and attendance tracking in the system. The Branch Manager assigns employee shifts, the system checks for conflicts, saves the schedule, and notifies employees.



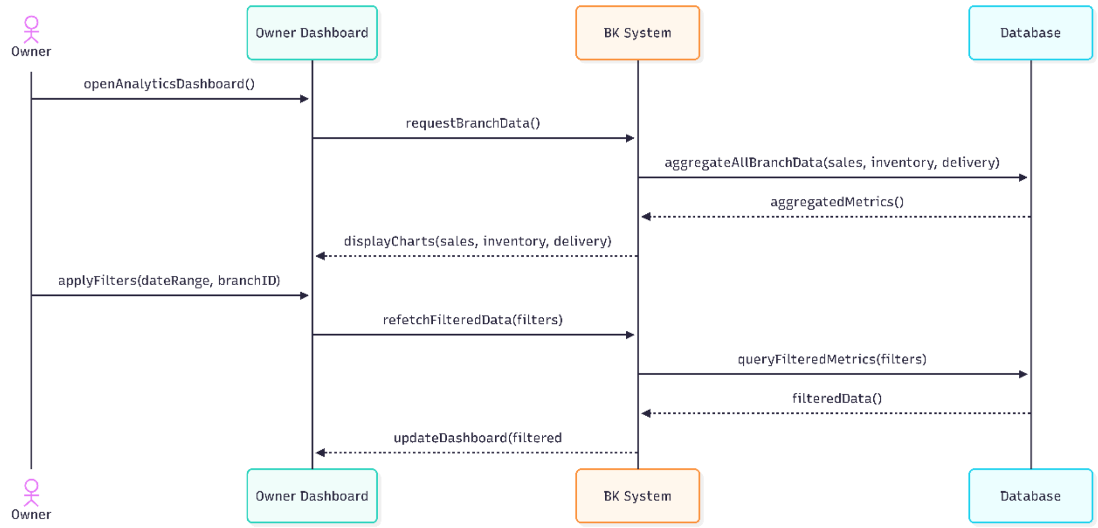
SSD-09 — Review Low-Stock Alerts & Process Stock Requests

This diagram illustrates the low-stock detection and replenishment process in the system. When inventory drops below set thresholds, the system alerts the Branch Manager, who creates a stock request that is sent to the Supplier. After approval and dispatch, inventory is updated and the Manager is notified once the request is fulfilled.



SSD-10— Monitor Centralized Branch Operations & Analytics

This diagram shows the Owner's analytics dashboard, which aggregates data from all branches, including sales, inventory, and delivery metrics. The system retrieves and displays this data as charts and tables, and allows filtering by date or branch to refresh the results for better decision-making.



4. Functional Requirements

BK Restaurant Management System — Functional Requirements			
ID	**Requirement**	**Description**	**Priority**
FR-01	Centralized Order Management	The system shall capture and store all online and walk-in customer orders in a centralized database, replacing the current WhatsApp and paper-based order management process.	**High**
FR-02	Digital Billing and Invoice Generation	The system shall automatically calculate order totals and generate digital bills and invoices through the Point of Sale (POS) system.	**High**
FR-03	Real-Time Kitchen Order Display	The system shall display customer orders on a digital kitchen screen in real time to reduce communication errors between reception staff and kitchen staff.	**High**
FR-04	Order Modification and Cancellation	The system shall allow authorized staff to modify or cancel customer orders before order completion and update all related system records accordingly.	**High**
FR-05	Delivery Assignment Management	The system shall allow restaurant staff to assign delivery orders to available riders through the system.	**High**
FR-06	Rider Mobile Application	The system shall provide a mobile application for delivery riders to receive delivery assignments and view customer location details.	**High**
FR-07	GPS-Based Delivery Tracking	The system shall provide real-time GPS tracking of delivery riders and display delivery progress on the owner dashboard.	**High**
FR-08	Delivery Status Updates	The system shall allow riders to update delivery statuses such as Assigned, Dispatched, In Progress, and Delivered.	**High**
FR-09	Inventory Management	The system shall automatically update inventory quantities after each completed sale.	**High**
FR-10	Low Stock Alert Generation	The system shall generate automatic alerts when stock levels fall below predefined minimum thresholds.	**High**
FR-11	Supplier Request Management	The system shall allow branch managers to create and send digital inventory requests to suppliers and maintain request records.	**Medium**

FR-12	Employee Attendance Management	The system shall record employee attendance digitally and maintain attendance history.	**Medium**
FR-13	Leave Request Management	The system shall allow employees to submit leave requests and allow managers to approve or reject requests through the system.	**Medium**
FR-14	Employee Schedule Management	The system shall allow managers to create, update, and monitor employee work schedules.	**Medium**
FR-15	Multi-Branch Monitoring Dashboard	The system shall provide a centralized dashboard that allows the restaurant owner to monitor all branches simultaneously.	**High**
FR-16	Sales and Revenue Reporting	The system shall generate daily, weekly, and monthly sales reports for each branch and for all branches combined.	**High**
FR-17	Role-Based Access Control	The system shall enforce role-based access control for Restaurant Owners, Branch Managers, Reception Staff, Kitchen Staff, and Delivery Riders.	**High**
FR-18	Secure User Authentication	The system shall require users to authenticate themselves before accessing system functions.	**High**

5. Non-Functional Requirements

BK Restaurant Management System — Non-Functional Requirements			
(i) Performance			
ID	**Category**	**Description**	**Priority**
NFR-01	Response Time	The system shall display POS order updates within 2 seconds.	**High**
NFR-02	Concurrent Access	The system shall support concurrent access from all restaurant branches without significant performance degradation.	**High**
NFR-03	Real-Time Processing	The system shall process inventory updates and delivery status updates in real time.	**High**
(ii) Security			
ID	**Category**	**Description**	**Priority**
NFR-04	User Authentication	The system shall require secure user authentication for all users.	**High**
NFR-05	Password Encryption	Passwords shall be stored using secure encryption techniques.	**High**
NFR-06	Role-Based Restriction	The system shall restrict access to features based on user roles.	**High**
NFR-07	Audit Logs	The system shall maintain audit logs of important user activities.	**Medium**
(iii) Usability			
ID	**Category**	**Description**	**Priority**
NFR-08	Simple UI	The user interface shall be simple and easy to use for non-technical staff.	**High**
NFR-09	Clear Navigation	The system shall provide clear navigation and user-friendly screens.	**High**
NFR-10	Rider App Usability	The rider mobile application shall be easy to learn and operate with minimal training.	**High**
(iv) Reliability			
ID	**Category**	**Description**	**Priority**
NFR-11	Offline Sync	The system shall prevent data loss during network failures through offline synchronization mechanisms.	**High**

NFR-12	Data Consistency	The system shall maintain accurate and consistent records across all branches.	**High**
NFR-13	Backup & Recovery	The system shall provide backup and recovery mechanisms for critical business data.	**High**
(v) Scalability			
ID	**Category**	**Description**	**Priority**
NFR-14	Branch Expansion	The system shall support the addition of new restaurant branches without major architectural changes.	**Medium**
NFR-15	Load Scalability	The system shall support increasing numbers of users, orders, and inventory records while maintaining acceptable performance.	**Medium**
(vi) Maintainability			
ID	**Category**	**Description**	**Priority**
NFR-16	Modular Architecture	The system shall use a modular architecture to simplify future enhancements and maintenance.	**Medium**
NFR-17	Coding Standards	The system shall follow standard coding practices and documentation guidelines.	**Medium**
NFR-18	Non-Disruptive Updates	The system shall allow software updates without significant disruption to restaurant operations.	**Medium**
(vii) Compatibility			
ID	**Category**	**Description**	**Priority**
NFR-19	Browser Support	The web application shall support modern web browsers including Google Chrome, Microsoft Edge, and Mozilla Firefox.	**High**
NFR-20	Android Support	The rider mobile application shall support Android smartphones with GPS capabilities.	**High**
(viii) Availability			
ID	**Category**	**Description**	**Priority**
NFR-21	Operating Hours	The system shall be available during restaurant operating hours with minimal downtime.	**High**
NFR-22	Continuous Operation	The system shall support continuous operation across all restaurant branches.	**High**

6. Alternative Solutions & Feasibility Study

6.1 Alternative Solutions Analysed

Alternative 1 – Manual / Semi-Digital System (Current Approach)

Aspect	Details
Description	Uses WhatsApp for orders, phone calls for delivery, and paper/spreadsheet-based records for billing, inventory, and employee management.
Advantages	No implementation cost, simple to use, no technical setup required.
Disadvantages	High chance of errors, missed orders, no real-time tracking, inefficient reporting, and poor scalability across multiple branches.
Cost	Very low initial cost but high long-term operational inefficiency cost.
Overall Assessment	Not suitable for scaling or improving efficiency.

Alternative 2 – Cloud-Based SaaS POS Solutions (e.g., Toast POS, Revel Systems)

Aspect	Details
Description	Utilize an existing cloud-based restaurant management and POS platform such as Toast POS or Revel Systems for order processing, billing, inventory management, and reporting.
Advantages	Quick deployment, vendor support, proven reliability, built-in reporting features, and reduced development effort.
Disadvantages	High recurring subscription costs, limited customization, dependency on third-party vendors, and some advanced features may require additional paid modules.
Cost	Ongoing monthly or annual subscription fees, plus potential costs for additional terminals and premium features.
Overall Assessment	Suitable for established businesses with larger budgets, but less cost-effective and flexible compared to a custom-built solution tailored to BK Restaurant's specific requirements.

Alternative 3 – Proposed Custom Web-Based System (BK Restaurant System)

Aspect	Details
Description	A custom-built system using React.js (frontend), Node.js (backend), and MySQL/PostgreSQL for database with modules for POS, inventory, orders, and reporting.
Advantages	Fully customized to business needs, supports multi-branch operations, reduces manual work, improves accuracy, and enables future expansion.
Disadvantages	Requires development time and maintenance responsibility.
Cost	One-time development cost & minimal hosting/maintenance cost.
Overall Assessment	**Recommended solution** as it best fits the business requirements and solves identified operational issues.

6.2 Feasibility Study

6.2.1 Technical Feasibility

All required technologies are mature, well-documented, and proven in production restaurant and delivery management systems:

Factor	Analysis
Frontend Technologies	React.js and Tailwind CSS are mature, widely adopted technologies suitable for developing responsive web interfaces and POS dashboards.
Backend Technologies	Node.js and Express provide a scalable and efficient platform for handling orders, inventory, reporting, and API services.
Database Technologies	PostgreSQL offers reliable relational data management, while Redis can be used for caching and performance optimization.
Integration Support	The system can support integrations such as thermal receipt printers and Google Maps services for delivery tracking and navigation.
Conclusion	The proposed system is technically feasible as the required technologies are well-established and suitable for restaurant management operations.

6.2.2 Economic Feasibility

Factor	Analysis
Development Cost	The system will be developed as part of the project using existing team resources, reducing software acquisition costs.
Software Cost	Most development tools and frameworks are open-source and free to use.

Factor	Analysis
Hosting Cost	Cost-effective hosting options such as AWS Free Tier or Railway.app can be used during deployment.
Long-Term Benefits	The system reduces manual work, minimizes operational errors, and eliminates recurring subscription fees associated with commercial SaaS solutions.
Conclusion	The project is economically feasible due to its low implementation cost and potential operational savings.

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6.2.3 Operational Feasibility

Factor	Analysis
User Readiness	Restaurant staff and management possess basic computer and smartphone literacy required to operate the system.
Ease of Use	The system is designed with a simple and intuitive user interface to minimize training requirements.
Business Compatibility	The proposed solution aligns with existing restaurant workflows while improving efficiency through automation.
User Acceptance	The client has expressed interest in a centralized solution to address current operational challenges.
Conclusion	The system is operationally feasible and can be adopted with minimal disruption to current business processes.

The system is designed for low-technical-proficiency users across all three branches:

- **POS (Reception/Kitchen):** Large touch targets, visual KDS display, minimal typing. New staff require no more than 30 minutes of training.
- **Rider Mobile App:** Step-by-step guided delivery flow, integrated navigation, simple status update buttons. Designed for riders with basic smartphone literacy.
- **Owner Dashboard:** Standard web conventions. Single-page analytics with pre-built charts requiring no manual data entry.
- **Training Plan:** On-site training sessions planned at the Ambalangoda branch for staff rollout before live deployment. BK branches will serve as UAT testing locations.

6.2.4 Schedule Feasibility

Phase	Duration	Timeline	Key Deliverables
Phase 1	1 week	Weeks 1–4	Idea discovery, problem identification, initial research
Phase 2	1 week	Weeks 1-2	System design, architecture diagrams, database schema, UI wireframes
Phase 3	1 week	Weeks 2	Project kick-off, team role allocation, repository setup
Phase 4	2 weeks	Weeks 3-4	Requirements engineering, stakeholder interviews, competitor analysis, literature review
Phase 5	1 weeks	Weeks 5	SRS preparation, requirements validation, SRS finalization
Phase 6	2 weeks	Weeks 5-7	System design, architecture design, database schema, use case diagrams, wireframes
Phase 7	1 week	Weeks 7–8	Design review, documentation consolidation, final submission preparation
Phase 8	4 Months	Month 2- 6	System development, implementation, testing, UI refinement, deployment preparation
Total	Approximately 6 Months	Approximately 2 Semesters	Complete restaurant management system delivery

Conclusion: The project schedule is feasible as the work is divided into manageable phases aligned with the academic timeline. The sprint-based approach allows requirements analysis, design, and development activities to be completed within the allocated project duration.

6.3 Justified Recommendation

The **custom web-based restaurant management system (BK Restaurant System)** is the most suitable solution because it directly addresses the operational problems of the current system while remaining cost-effective and scalable.

Unlike manual processes, it eliminates errors and improves efficiency. Compared to SaaS solutions, it avoids high recurring costs and provides full customization for local restaurant workflows, including multi-branch operations.

Therefore, the proposed system is recommended as it offers the best balance between cost, usability, and functionality for the BK Restaurant environment.